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-----French-----

Exercice1:

- 1). Ecrire une fonction récursive *convertir* qui permet de convertir un nombre décimal en binaire.
- 2). Allouer dynamiquement la mémoire pour le tableau renvoyant le résultat.
- 3). Ecrire un programme principal de test.

-----English-----

Exercise 1:

- 1). Write a recursive function *convert* which allows to convert a decimal number to binary.
- 2). Dynamically allocate memory for array returning the result.
- 3). Write a main test program.

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#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
void get_binary(int decimal_x);
int main()
{
    char c;
    int decimal_x;
    printf("\n\t***** Converting Decimal numbers to binary*****\n");
    do
    {
        printf("Number = ");
        scanf("%d", &decimal_x);
        printf("Binary of %d is ", decimal_x);
        get_binary(decimal_x);
        printf("\n\nDo you want to enter another number to convert it into decimal?
(y/n): ");
        c = getch();
        fflush(stdin);
        printf("\n");
    } while (c == 'y' || c == 'Y');
    return 0;
}

void get_binary(int decimal_x)
{
    int i = 0, length = 1, *binary;
    binary = (int *)malloc(length * sizeof(int));
    if (decimal_x != 0)
    {
        *(binary + i) = decimal_x % 2;
        decimal_x = decimal_x / 2;
        get_binary(decimal_x);
        length++;
        printf("%d", *(binary + i));
        i++;
    }
}

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